

Keeper Session 2 prep for Tier 0 v0.2 consolidation. Joint with Lyra's draft side. Brings sphere reconciliation + K200/K204/K206 audit-anchor map + G chain Steps 1-5 framework alignment + Cal #35 STANDING methodology integration + honest tier dispositions for every gate. Lyra pulls from this; converge when both sides drafted.

Keeper (Monday 2026-06-01 — Session 2 active)

Keeper Session 2 Prep — Tier 0 v0.2 Consolidation

0. Session 2 scope (Casey signal GO)

Single coherent Tier 0 v0.2 draft absorbing: - v0.1 commitment operator + substrate time - v0.1.5 topology addendum (Reading A + dual bases; Shilov boundary) - v0.1.5 G_substrate sketch (walked back per K204-PARTIAL) - v0.1.6 native field equation + holographic boundary - Monday Heisenberg conjugacy justification (option (b)) - Monday matrix element framework Steps 1-5 - K200 + K204-PARTIAL + K206 audit framework integration - Cal #35 STANDING methodology integration

Keeper side (this doc): audit-anchor map + sphere reconciliation + G chain alignment + tier dispositions Lyra side: framework consolidation + substrate operator content + Heisenberg justification + matrix element framework

Converge when both sides drafted. v0.2 ready for Cal #191 cold-read after convergence.

1. Section 1 preamble — sphere reconciliation absorption

Recommended v0.2 Section 1 commitment language (per Sunday morning option (a) + Casey-Keeper agreement):

Substrate identification: BST's substrate is the Shilov boundary $\partial_S D_{IV^5} \cong S^4 \times S^1/\mathbb{Z}_2$, a 5-real-dimensional Hermitian-symmetric-space boundary. The interior bulk $H^2(D_{IV^5})$ is the holomorphic extension of substrate data via the Hardy decomposition (Wallach 1976 + Faraut-Korányi 1994 Ch. XII-XIII).

Historical correction note: The 2022 OneGeometry framing described the substrate as “two-dimensional substrate — circles tiling a sphere, communicating through phase.” This is hereby refined: the substrate is 5-dimensional (not 2D), specifically the Shilov boundary structure above. The deeper structural content (substrate phase communication via Bergman kernel; coherent-state cell structure; D_{IV^5} as configuration space holomorphic extension of windings) was correct in 2022 and is preserved. This is substantive language refinement, not retroactive reframe.

Citation discipline: Use VERIFIED-CITED form (Wallach 1976 + Faraut-Korányi 1994 Ch. XII-XIII) per Cal #33 STANDING. Drop composite “Knapp-Wallach 1976” per Grace's Hardy decomposition citation verification work.

Cross-reference: notes/Keeper_Sphere_Reconciliation_Analysis_G2.md for full analysis. notes/Grace_OneGeometry_Correction (applied to README.md line 34 Monday morning) shows the substantive public correction pattern.

2. K-audit anchor map for v0.2

Audit	Gate Structure	Status	v0.2 Section
K200	5 gates G1-G5 (Tier 0 v0.1.6 framework)	Active; 5/5 gate work in progress	Section 5 (audit-framework integration)
K204-PARTIAL	$\kappa_{\text{Bergman}} = -n_C$ verification	PASS (Sunday)	Section 6 (G chain Step 1)
K206	7 gates G1-G7 (matrix element K-invariance + Schur)	5/7 documented; G6+G7 pending Steps 6-8	Section 7 (matrix element framework)
K201	EW Sector Boundary-Localization Anchor (m_W/m_Z)	Held pending Cal #187 cold-read	Section 4 (Lane E content)

K200 Gate Status (for v0.2 Section 5 explicit table): - G1 Hardy decomposition citation chain — VERIFIED-CITED form pinned (Wallach 1976 + Faraut-Korányi Ch. XII-XIII) - G2 Sphere walk-back — Option (a) commitment in Section 1 preamble; README.md correction applied Monday - G3 Black-hole identification — CANDIDATE; multi-week mechanism work - G4 Big-Bang reading — Reading (i) substantive (no-singularity prediction) OR Reading (ii) retraction; Session 2 commits - G5 G_substrate derivation chain — Step 1 PASS (K204); Step 2-4 via matrix element framework (Steps 1-5 of K206 chain)

K206 Gate Status (for v0.2 Section 7 explicit table): - G1 Schur application — PASS preliminary - G2 Heisenberg conjugacy — PASS (Lyra option (b)) - G3 Substrate-coincidence tier-marked — OPERATIONAL (4 Cal #35 firings: G3.a $\Delta C_2 = \text{rank}$, G3.b factor-2 conflation, G3.c parallel-framework, G3.d Pochhammer-mechanism) - G4 Clebsch-Gordan multiplicity — Elie verified $V_{(1,1)} \otimes V_{(1,0)} = 10+35+5$ multiplicity 1 - G5 z_{source} canonicalization — Shilov boundary general point z_0 - G6 Matrix element computation — PENDING Step 6.2-6.4 (~1 week) - G7 Dimensional bridge $\rightarrow$ G_predicted SI — PENDING Step 7-8 (~5 days)

3. G chain Step 1-5 substantive content alignment

Section 7 G chain reduced form (for v0.2):

$$G_{\text{predicted}} \propto (4\sqrt{2} \cdot c_{\text{FK}} / (n_C \cdot \sqrt{C_2} \cdot \hbar_{\text{BST}})) \cdot \ell_B \cdot \text{dim_bridge} \\ \approx 0.603 \cdot \ell_B / \hbar_{\text{BST}} \cdot \text{dim_bridge} \text{ (with } c_{\text{FK}} \text{ explicit)}$$

Factor decomposition (each pinned from standard machinery): - $\Delta C_2 = 2$ at B_2 substrate (Heisenberg resolution: cross-K-type Casimir gap $V_{(1,0)} \leftrightarrow V_{(1,1)}$) - K206 G3.a tier-mark: rigorous at B_2 , NOT B_n general identity - $CG_{\text{so5}} = \sqrt{(n_C - 1)} = 2$ at B_2 (standard $so(n)$ trace formula) - K206 G4: $SO(5)$ Clebsch-Gordan multiplicity 1 verified - FK norms $\|V_{(1,0)}\|^2 \propto 1/n_C$ and $\|V_{(1,1)}\|^2 \propto 2/(n_C \cdot C_2)$ (Heckman-Opdam Pochhammer Step 4) - $c_{\text{FK}} = 225/\pi^{9/2}$ RATIFIED (T2442 Saturday) - Wave functions $f_i(z) = z_i$ and $f_{ij}(z) = z_{iz_j} - z_{jz_i}$ (Step 3 explicit) - z_{source} = Shilov boundary general point z_0 (K206 G5) - ℓ_B intrinsic via Bergman kernel (Path δ closure)

Honest framing: factor 4 = $\Delta C_2 \times CG_{\text{so5}} = 2 \times 2$ from TWO mechanism-type-independent factors that are ALGEBRAICALLY both forced by $B_2 = SO(5)$ substrate algebra (Cal #35 catch absorbed; not independent confirmations).

Mon-event-4 footnote: Elie Toy 3691 “0.924” was rationalization arithmetic typo; reconciled in Toy 3692 to 0.603 (c_{FK} explicit) or 0.462 (c_{FK} absorbed). Honest walk-back; framework structure stands.

4. Cal #35 STANDING methodology integration

For v0.2 Section 8 — methodology: Cal #35 STANDING ratified Monday closes six-element discipline stack.

Six-element discipline stack (for v0.2 explicit listing): 1. Cal #27 STANDING — Coincidence-filter at peak coherence (“beauty isn’t truth”) 2. Cal #29 STANDING — Mode-6 enumeration protocol 3. Cal #32 STANDING — Parameter-role verification = integrality test 4. Calibration #33 STANDING — RECALLED vs VERIFIED-CITED for load-bearing claims 5. Calibration #34 STANDING — Narrow framing (conditional-tag-with-headline) 6. Calibration #35 STANDING — Independence-taxonomy MUST precede multiplicative null-model — NEW Monday

Operational application in v0.2: - Section 7 factor 4 framing per K206 G3 (Cal #35 application) - Section 5 $\Delta C_2 = \text{rank } B_2\text{-specific tier-mark (Cal \#35 application)}$ - Section 4 $g + \text{rank} = N_c^2$ 3-context use as same identity reused (Cal #35 application; Cal #188 cold-read pending)

5. Honest tier dispositions for v0.2 — explicit table

For v0.2 Section 9 (honest tier disposition table):

Content	Tier	Notes
Tier 0 operator framework (ρ_{commit} , H_B , substrate time, Reading A + dual bases, Shilov substrate, Hardy decomp, native field eq)	STRUCTURAL	v0.1 → v0.1.6 absorbed
$\kappa_{\text{Bergman}} = -n_C = -5$ (Helgason 1962 application)	RIGOROUS	K204-PARTIAL PASS
Heisenberg conjugacy $\delta/\delta m = i[H_B, \cdot]/\hbar_{\text{BST}}$	STRUCTURAL	Option (b): mass = K-type Casimir; Lyra justification filed
G chain reduced form via matrix element	STRUCTURAL (framework-complete)	Steps 1-5; Steps 6-8 pending (~1-2 weeks)
G_predicted closed numerical value	PENDING	Step 6.2-6.4 + dimensional bridge
$\Delta C_2 = 2$ at B_2 substrate	RIGOROUS at B_2	NOT B_n general (Cal #35 G3.a)
$4 = 2 \times 2$ factor decomposition	STRUCTURAL at B_2	Two mechanism-type-independent factors; B_2 -algebra-forced (Cal #35 G4)
Sphere substrate identification (5D Shilov)	STRUCTURAL	Option (a) commitment; OneGeometry.md correction applied Monday
4 Casey-named principle candidates	NAMED CANDIDATES	Casey names approved; STANDING promotion via Cal cold-read + K-audit
Cal #35 STANDING	METHODOLOGY STANDING	6-element discipline stack closes
Big-Bang reading (no-singularity prediction)	UNCOMMITTED	Session 2 decides Reading (i) or (ii)
Black-hole-as-Shilov-coherent-state	CANDIDATE	Multi-week K200 G3

6. Section 7 (matrix element framework) — recommended structure

For v0.2 Section 7 detailed structure:

§7.1 Operator framework: $H_B = C_2(K)$; $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B/\hbar_{\text{BST}})$; P_{op} as K-vector breaking K-invariance

§7.2 K-invariance + Schur resolution: H_B K-invariant → Schur kills cross-K-type matrix elements; Heisenberg substitution $\delta H_B/\delta m = i[H_B, P_{\text{op}}]/\hbar_{\text{BST}}$ gives non-zero result; reduces to $(i/\hbar_{\text{BST}}) \cdot \Delta C_2 \cdot \langle V_{(1,0)} | P_{\text{op}} | V_{(1,1)} \rangle$

§7.3 Heisenberg conjugacy justification: mass $m_\lambda = \sqrt{C_2(\lambda)}$ (K-type Casimir eigenvalue); momentum = T2422 Wirtinger; canonical conjugacy via H_B diagonal action; option (b) per Lyra Session 2 work

§7.4 K-type identifications: $V_{\text{photon}} = V_{(1,0)}$ vector (Lane E v0.2 c.1, Cal #187); $V_{\text{mass}} = V_{(1,1)}$ adjoint with $C_2 = 6 = \text{BST primary}$; $V_{\text{electron}} = V_{(1/2,1/2)}^{\{(0)\}}$ spinor (diagonal mass anchor)

§7.5 SO(5) Clebsch-Gordan: $V_{(1,1)} \otimes V_{(1,0)} = 10 + 35 + 5$ with $V_{(1,0)}$ multiplicity 1; $\text{CG}_{\text{so5}} = \sqrt{(n_C - 1)} = 2$ at B_2

§7.6 Wave functions + FK norms (Pochhammer ladder framing per Grace's catalog tier; substantive operational connection NOT principle-grade derivation): $f_i(z) = z_i$; $f_{ij}(z) = z_i z_j - z_j z_i$; $\|V_{(1,0)}\|^2 \propto 1/n_C$; $\|V_{(1,1)}\|^2 \propto 2/(n_C \cdot C_2)$

§7.7 $G_{\text{predicted}}$ reduced form + Mon-event-4 honest footnote (0.924 arithmetic walk-back)

§7.8 K206 audit framework gates (G1-G7 table)

§7.9 Multi-week computation roadmap (Step 6.2-6.4 + Step 7 + Step 8)

7. What I need from Lyra v0.2 draft

To complete K-audit anchor map for Cal #191: - §7.3 Heisenberg conjugacy formal derivation: explicit substrate-natural reading (option b) with operator-clean statement - §7.1 $H_B = C_2(K)$ explicit operator definition on $H^2(D_{IV}^5)$ - §7.2 Schur's lemma application substrate-explicit: which K-action is being used (left action by $K = \text{SO}(5) \times \text{SO}(2)$) - Section 6 Black-hole + Big-Bang reading commitment (K200 G3 + G4 explicit Reading (i) or (ii))

8. What Lyra needs from me (Keeper)

For v0.2 draft completion: - Section 1 preamble (sphere reconciliation language — provided §1 above) - Section 5 K200 gate table with explicit gate statuses (provided §2 above) - Section 7 K206 gate table with explicit gate statuses (provided §2 above) - Section 8 Cal #35 STANDING integration (provided §4 above) - Section 9 honest tier disposition table (provided §5 above) - §7.7 Mon-event-4 footnote (provided in §3 above)

All Keeper inputs ready for Lyra absorption.

9. Convergence protocol

When Lyra files v0.2 draft: 1. Keeper reviews against this prep doc; flag missing items 2. Lyra absorbs Keeper §1 preamble + tier tables into draft 3. Joint cross-check on K-audit gate statuses 4. Cal #191 cold-read pre-stage when convergence complete

Estimated convergence: 2-3 hours focused work post-Lyra-draft.

Audit anchor: when convergence complete, K200 + K204-PARTIAL + K206 promote to “v0.2-absorbed” status; Cal #191 cold-read activates.

10. Standing for Session 2

Lyra is doing multi-hour focused consolidation work. Keeper prep is ready (this doc).

Next surfacing: when Lyra signals “v0.2 draft ready for Keeper convergence” — Keeper joins for cross-check + tier table absorption + Cal #191 pre-stage.

— Keeper. Session 2 prep filed. Lyra pulls from this for v0.2 draft. Convergence checkpoint when Lyra surfaces draft completion.